

A. Toy Marbles

2 seconds, 256 megabytes

Busy Beaver has discovered that someone has mixed up his toy marbles!



There are N containers, numbered from 1 to N . The i -th container currently contains a single marble of color c_i .

Busy Beaver wants to tidy up his marbles so that the i -th container only contains marbles of color i . To achieve this, he can perform either of the following actions any number of times (possibly zero):

- Swap the marbles in two containers x and y . After this action, all marbles from container x move to container y , and vice versa.
- Move all the marbles from one container y to another container x . After this action, container y becomes empty, and all its marbles are moved to container x .

Find a way to organize the marbles using the *minimum* number of actions.

Input

The first line contains an integer N ($1 \leq N \leq 2 \cdot 10^5$) — the number of containers.

The second line contains N integers $c_1, c_2, \dots, c_N$ ($1 \leq c_i \leq N$) — the marble initially in each container.

Output

On the first line, print a single integer K — the minimum number of actions required.

On the next K lines, describe the actions in order, one per line. Each action should be in one of the following formats:

- 1 x y : Swap the marbles in containers x and y ($1 \leq x, y \leq N$; $x \neq y$).
- 2 x y : Move all marbles from container y to container x ($1 \leq x, y \leq N$; $x \neq y$).

If there are multiple ways to achieve the goal in the minimum number of actions, you may print any valid solution.

Scoring

There are two subtasks for this problem.

- (20 points) All c_i are distinct.
- (80 points) No additional constraints.

input
4
2 4 3 1
output
2
1 2 4
1 1 2

input
8
3 6 7 6 3 6 8 3
output
6
2 6 4
2 6 2
1 8 7
1 7 3
2 3 1
2 3 5

In the first test case, it is able to achieve the goal in two actions:

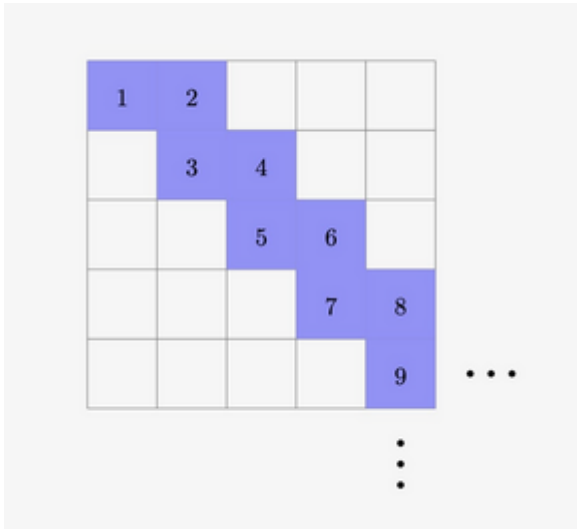
- Swap the marbles in containers 2 and 4.
- Swap the marbles in containers 1 and 2.

It is impossible to achieve the goal in less than two actions.

B. Snakes on a Grid

3 seconds, 256 megabytes

On a rectangular grid, a *snake* of size k is defined to be a subset of cells that is formed by taking the first k numbered cells in the following figure, where cells are numbered along a path with alternating right and down steps:



Additionally, subsets that are **rotations, translations, or reflections of these shapes are also considered snakes**. A grid is considered *good* if every connected component consisting of equal values in the grid is a snake.

Busy Beaver has an $N \times M$ grid with rows numbered $0, \dots, N - 1$ and columns numbered $0, \dots, M - 1$, where each cell (i, j) in row i , column j has a value $a_{i,j}$. Since he's afraid of snakes, he wants you to answer Q of the following queries: given a contiguous subrectangle of the grid, determine whether or not the subgrid is good.

Input

The first line contains 2 integers, N and M ($1 \leq N, M \leq 1000$) — the dimensions of the grid.

Each of the next N rows contains M integers, with the j -th element of the i -th row being $a_{i,j}$ ($1 \leq a_{i,j} \leq 10^6$) — the number written in the j -th cell of the i -th row.

The next line contains a single integer, Q ($1 \leq Q \leq 5 \cdot 10^5$) — the number of queries.

The next Q lines describe the queries, Each line contains 4 integers x_1, y_1, x_2, y_2 ($0 \leq x_1 \leq x_2 < N, 0 \leq y_1 \leq y_2 < M$), representing a query asking whether the rectangular subgrid with top left corner at (x_1, y_1) and bottom right corner at (x_2, y_2) is good.

Output

For each query, output "YES" (without quotes) if the corresponding subgrid consists only of snakes, and "NO" (without quotes) otherwise.

You can output "YES" and "NO" in any case (for example, strings "yES", "yes" and "Yes" will be recognized as a positive response).

Scoring

There are four subtasks for this problem.

- (15 Points): The sum of the areas of all the queries does not exceed 10^6 .
- (15 Points): Every connected component of equal values in the grid has size at most 4.
- (30 Points): Every connected component of equal values in the grid has size at most 5.
- (40 Points): No additional constraints.

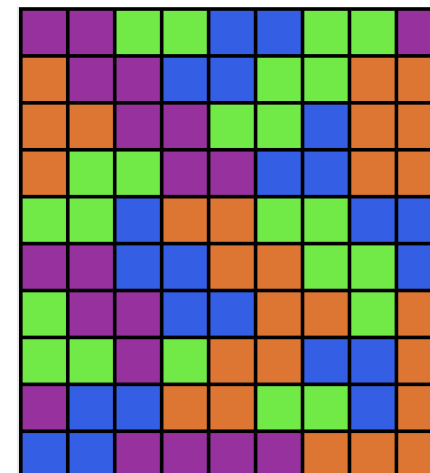
input

```
10 9
1 1 2 2 3 3 2 2 1
4 1 1 3 3 2 2 4 4
4 4 1 1 2 2 3 4 4
4 2 2 1 1 3 3 4 4
2 2 3 4 4 2 2 3 3
1 1 3 3 4 4 2 2 3
2 1 1 3 3 4 4 2 4
2 2 1 2 4 4 3 3 4
1 3 3 4 4 2 2 3 4
3 3 1 1 1 1 4 4 4
5
0 1 6 5
5 4 7 8
0 6 2 8
6 0 9 3
8 5 9 8
```

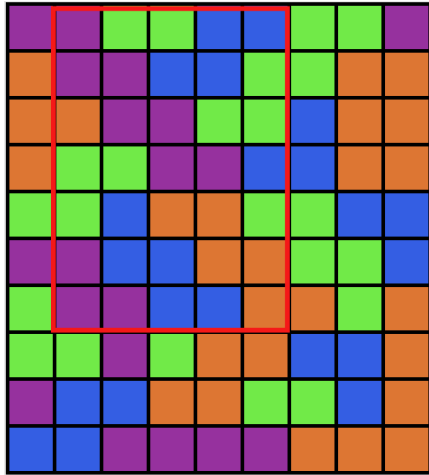
output

```
YES
NO
NO
YES
NO
```

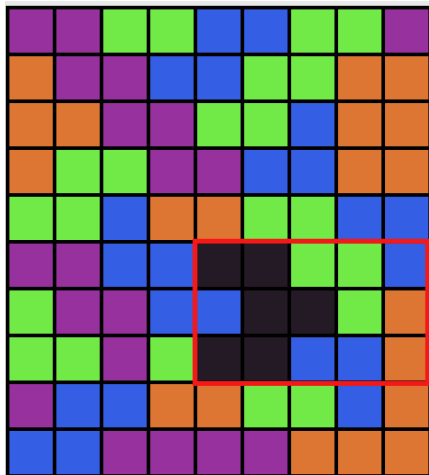
If we color the sample grid with respect to the value in each cell, we get



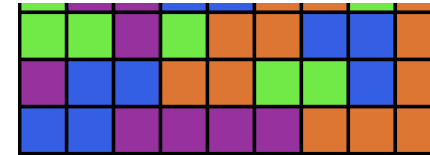
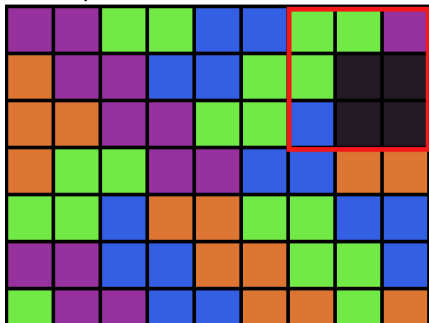
In the first query, all components are snakes.



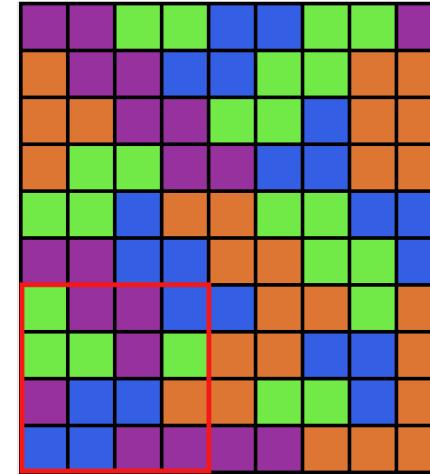
In the second query, the component that is not a snake is colored in black.



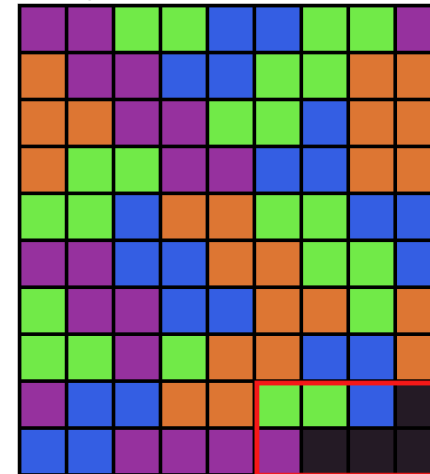
In the third query, the component that is not a snake is colored in black.



In the fourth query, all components are snakes.



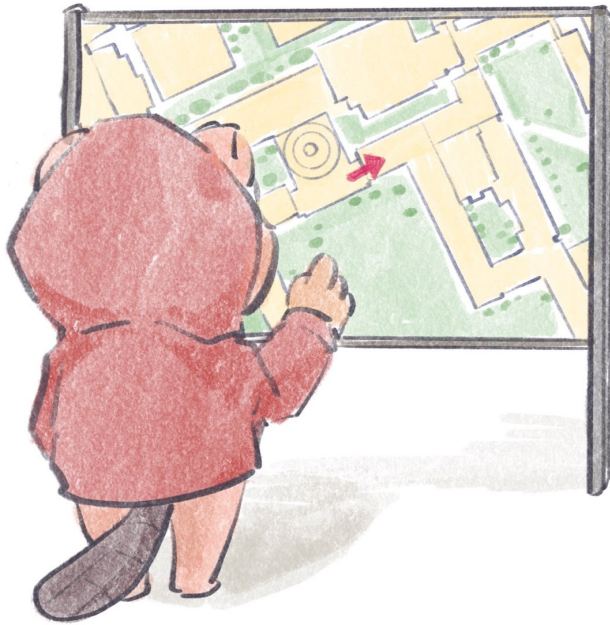
In the fifth query, the component that is not a snake is colored in black.



C. MIT Tour

3 seconds, 256 megabytes

Busy Beaver is visiting MIT, so he decided to have a tour around campus.



The campus consists of N rooms and $N - 1$ corridors between them. The rooms are enumerated consecutively from 1 to N , and the i -th corridor has a known length w_i and can be traversed in both directions. The layout of campus is also such that between any two rooms, there is exactly one way to go from one to another through these corridors without going through any room twice. Define the *level* of a room to be the minimum number of corridors needed to get from the Great Dome (room 1) to this room. For instance, the Great Dome is level 0, and any room directly connected to it would be level 1. Let k be the highest level of any room on campus, and denote $d(u, v)$ to be the length of the shortest path between rooms u, v .

Busy Beaver wants his tour to be interesting. Starting from the Great Dome (room 1), he wishes to visit a sequence of rooms $a_1, \dots, a_k$ such that for each $1 \leq i \leq k$, room a_i is on level i , and for each $1 \leq i < k$, rooms a_i, a_{i+1} are **not** connected by a corridor. As Busy Beaver is busy as usual, he also wants to minimize the total distance of the tour, which is

$$d(1, a_1) + d(a_1, a_2) + \dots + d(a_{k-1}, a_k).$$

Since the MIT campus is very complex, Busy Beaver needs your help to plan out his tour. Help him find out the minimum length of an interesting tour.

Input

The first line consists of a single integer N ($2 \leq N \leq 2 \cdot 10^5$) — the number of rooms at the MIT campus.

Each of the next $N - 1$ lines contains three integers u, v, w ($1 \leq u, v \leq N, u \neq v, 1 \leq w \leq 10^7$) — a corridor between rooms u and v with the length w .

It is guaranteed that there is exactly one simple path between any two rooms, and that at least one interesting tour exists.

Output

Output a single integer, the minimal distance of an interesting MIT tour.

Scoring

There are three subtasks for this problem.

- (15 points): $N \leq 10^3$.
- (25 points): $N \leq 10^5$ and the maximum level of any room satisfies $k \leq 100$.
- (60 points): No additional constraints.

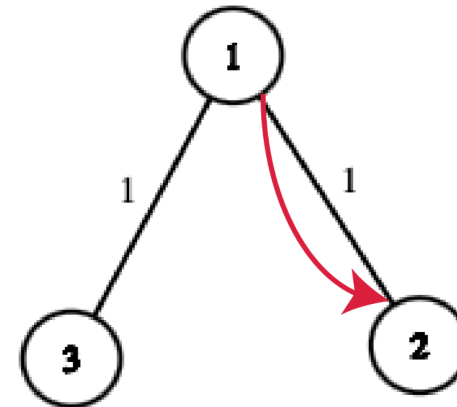
input
3 1 2 1 1 3 1
output
1

input
5 1 2 3 1 3 1 3 4 2 3 5 3
output
9

In the first test case, there are two possible interesting tours:

- $[a_1] = [2]$ — The total distance of the tour will be 1.
- $[a_1] = [3]$ — The total distance of the tour will be 1.

They both obtain the minimum possible length of 1, which is the answer for this case.

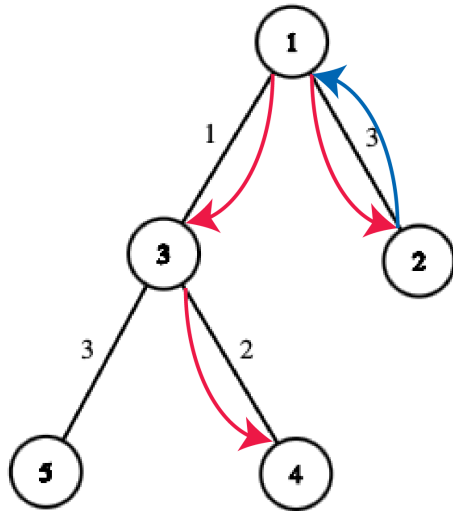


In the second test case, notice that both rooms of level 2 are connected to room 3. Therefore, no interesting tour can have $a_1 = 3$, or otherwise it will violate the condition that a_1, a_2 are not connected by a corridor.

Therefore, there are two possible interesting tours:

- $[a_1, a_2] = [2, 4]$ — The tour will traverse edges $1 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 4$. The total distance is $d(1, a_1) + d(a_1, a_2) = 3 + 6 = 9$.
- $[a_1, a_2] = [2, 5]$ — The tour will traverse edges $1 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 5$. The total distance is $d(1, a_1) + d(a_1, a_2) = 3 + 7 = 10$.

Therefore, the minimum length of an interesting tour in this case is 9.



D. Path Partition

15 seconds, 1024 megabytes

This is an "output-only" problem. The test cases are made available to you. However, due to the large output size, you should still submit a program that reads the input and produces the outputs within the time limit.

Busy Beaver generated a **random** undirected graph with N vertices and M edges, where M is a multiple of 3. Now, he wants to partition the edges into $M/3$ paths of length 3 (possibly starting and ending at the same vertex). Can you help Busy Beaver find such a partition?

Input

Please download the test data using [this link](#).

The first line of input contains two integers N and M — the number of vertices and edges in the graph, respectively.

The i th of the next M lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq N, u_i \neq v_i$) — the endpoints of the i th edge.

It is guaranteed that the M edges were generated randomly. Formally, out of the $\frac{N(N-1)}{2}$ possible edges, M distinct edges were chosen uniformly at random without replacement.

Output

Output $M/3$ lines. The i th line should contain 4 integers a_i, b_i, c_i , and d_i ($1 \leq a_i, b_i, c_i, d_i \leq N$), representing a path using edges (a_i, b_i) , (b_i, c_i) , and (c_i, d_i) .

It is allowed to have $a_i = d_i$. Your output should satisfy

$$\bigcup_{i=1}^{M/3} \{\{a_i, b_i\}, \{b_i, c_i\}, \{c_i, d_i\}\} = \bigcup_{i=1}^M \{\{u_i, v_i\}\}.$$

Scoring

There are 10 test cases in total, each worth 10 points. For your convenience, the values of N and M in each test case are shown in the following list:

- Test 1: $N = 771, M = 296835$.
- Test 2: $N = 772, M = 297606$.
- Test 3: $N = 1000, M = 300000$.
- Test 4: $N = 3000, M = 300000$.
- Test 5: $N = 10000, M = 300000$.
- Test 6: $N = 30000, M = 300000$.
- Test 7: $N = 50000, M = 300000$.
- Test 8: $N = 70000, M = 300000$.
- Test 9: $N = 80000, M = 300000$.

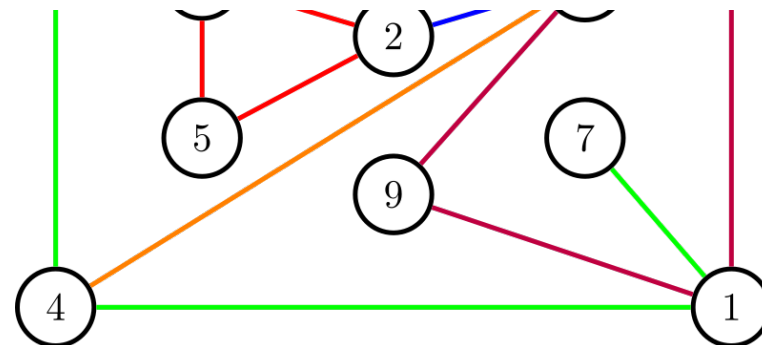
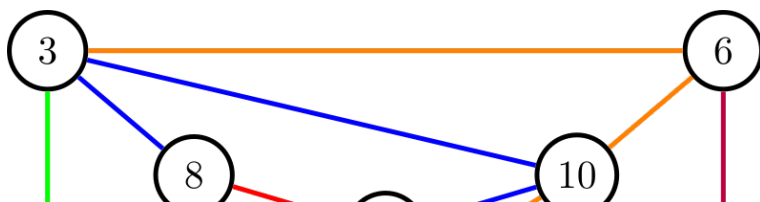
- Test 10: $N = 90000$, $M = 300000$.

It is guaranteed that each of the 10 generated graphs has a valid partition.

input	
10	15
3	10
10	4
3	6
5	8
10	9
5	2
1	4
1	9
1	7
8	3
1	6
6	10
10	2
8	2
4	3
output	
2	5 8 2
8	3 10 2
3	4 1 7
6	1 9 10
3	6 10 4

The sample is provided to illustrate the input and output format. It is not scored.

The sample graph and partition are shown in the figure below. Note that the path $2 - 5 - 8 - 2$ starts and ends at the same vertex, which is allowed.



E. Colored Blocks

1 second, 256 megabytes



Busy Beaver is playing with his favorite blocks. There are N blocks arranged in a row, each block having a color c_i . Busy Beaver hates seeing blocks of the same color next to each other, so he would like to rearrange the blocks into multiple rows, such that the blocks in each row are in the same ordering as they were in the original line, and no line has two consecutive blocks of the same color. Each of the N original blocks should belong to exactly one of these rows.

Please help Busy Beaver determine an arrangement of blocks that uses the least possible number of lines.

Input

Each test contains multiple test cases. The first line of input contains a single integer T ($1 \leq T \leq 10^5$), the number of test cases. The description of each test case follows.

The first line of each test case contains a single positive integer N ($1 \leq N \leq 2 \cdot 10^5$) — the number of blocks.

The second line contains N space separated integers c_i ($1 \leq c_i \leq 10^9$) — the color of the i th block.

It is guaranteed that the sum of N across all test cases is no more than $2 \cdot 10^5$.

Output

For each test case, the first line of output should contain a single positive integer k , the minimum number of rows that Busy Beaver needs to rearrange the blocks to satisfy his conditions.

For each of the next k lines, output the number of blocks in that row, as well as the indices of the blocks themselves, space separated. Each of the rows must be nonempty.

If there are multiple solutions with the same minimum number of rows needed, output any one of them.

Scoring

You will receive 30% of the points for each subtask if your value of k is correct, but the construction is incorrect. **To receive this partial credit, you still need to output k lines for each test describing some rearrangement of the blocks into k nonempty rows.**

For instance, the following output would receive partial credit on the sample input:

```
2
1 1
1 2
1
3 1 2 3
2
3 1 2 3
1 4
```

while the following does not:

```
2
2 1 2
0
1
0
2
1 1
1 2
```

There are two subtasks for this problem.

- (30 points): The sum of N across all test cases does not exceed $5 \cdot 10^3$.
- (70 points): No additional constraints.

input
3 2 2 2 3 1 2 1 4 1 1 2 1
output
2 1 1 1 2 1 3 1 2 3 2 1 1 3 2 3 4

In the first test case, it is optimal to create two rows, each with one of the two blocks. It is impossible to use only one row, since the two blocks have the same color.

In the second test case, we can use one row to contain all three blocks.

In the third test case, it is optimal to create two rows, with one containing the first block and the second containing the remaining three blocks.